

2020 AMC 10B Problem 3 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 3

Problem:

The ratio of w to x is $4 : 3$, the ratio of y to z is $3 : 2$, and the ratio of z to x is $1 : 6$. What is the ratio of w to y ?

Answer Choices:

- A. $4 : 3$
- B. $3 : 2$
- C. $8 : 3$
- D. $4 : 1$
- E. $16 : 3$

Solution:

We have

$$\frac{w}{y} = \frac{w}{x} \cdot \frac{x}{z} \cdot \frac{z}{y} = \frac{4}{3} \cdot \frac{6}{1} \cdot \frac{2}{3} = \frac{16}{3}$$

from which $w : y = \boxed{(E)16 : 3}$.

OR

We need to somehow link all three of the ratios together. We can start by connecting the last two ratios together by multiplying the last ratio by two.

$$z : x = 1 : 6 = 2 : 12$$

and since

$$y : z = 3 : 2$$

we can link them together to get

$$y : z : x = 3 : 2 : 12$$

Finally, since

$$x : w = 3 : 4 = 12 : 16$$

we can link this again to get:

$$y : z : x : w = 3 : 2 : 12 : 16$$

so

$$w : y = \boxed{(E)16 : 3}$$

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